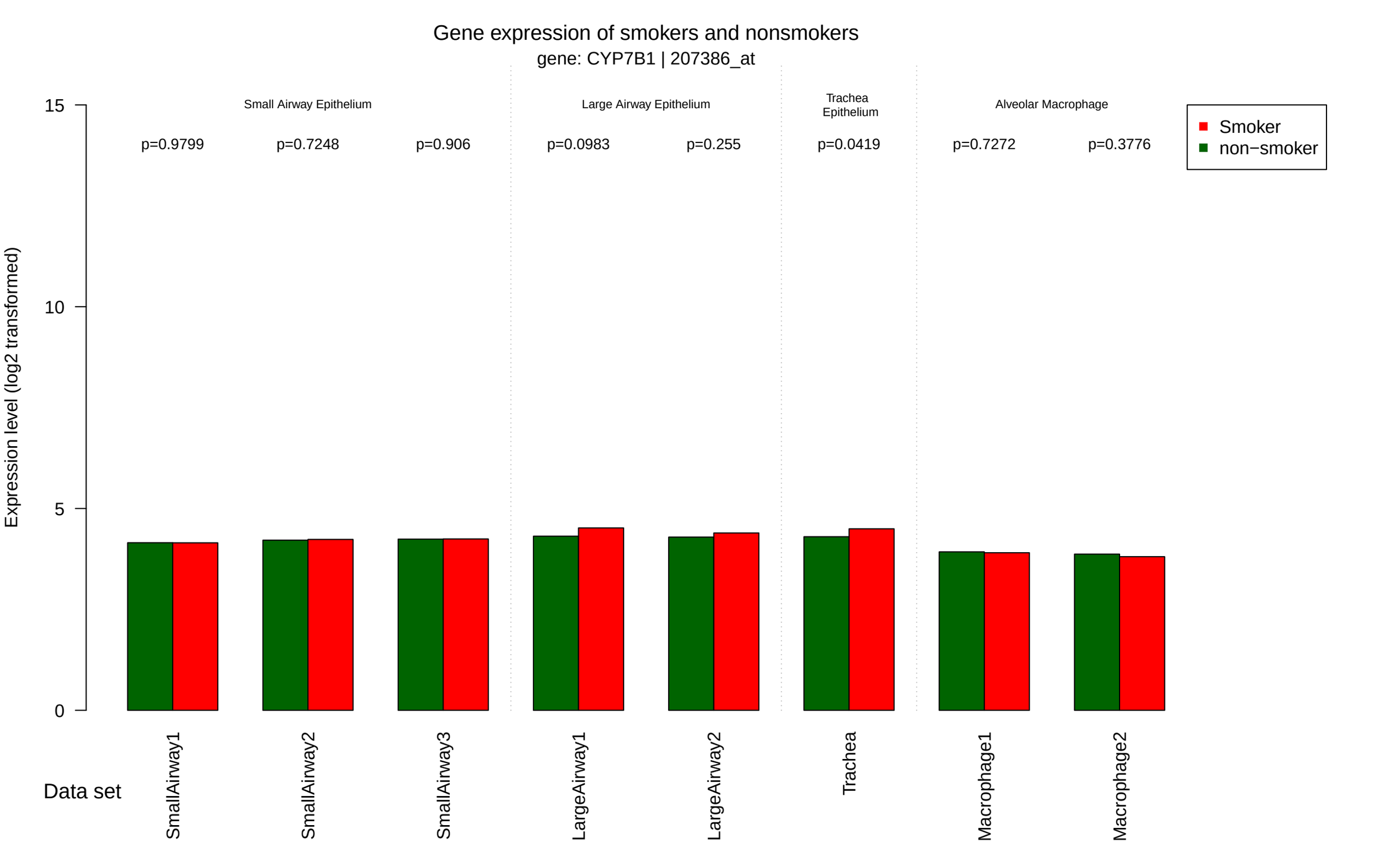
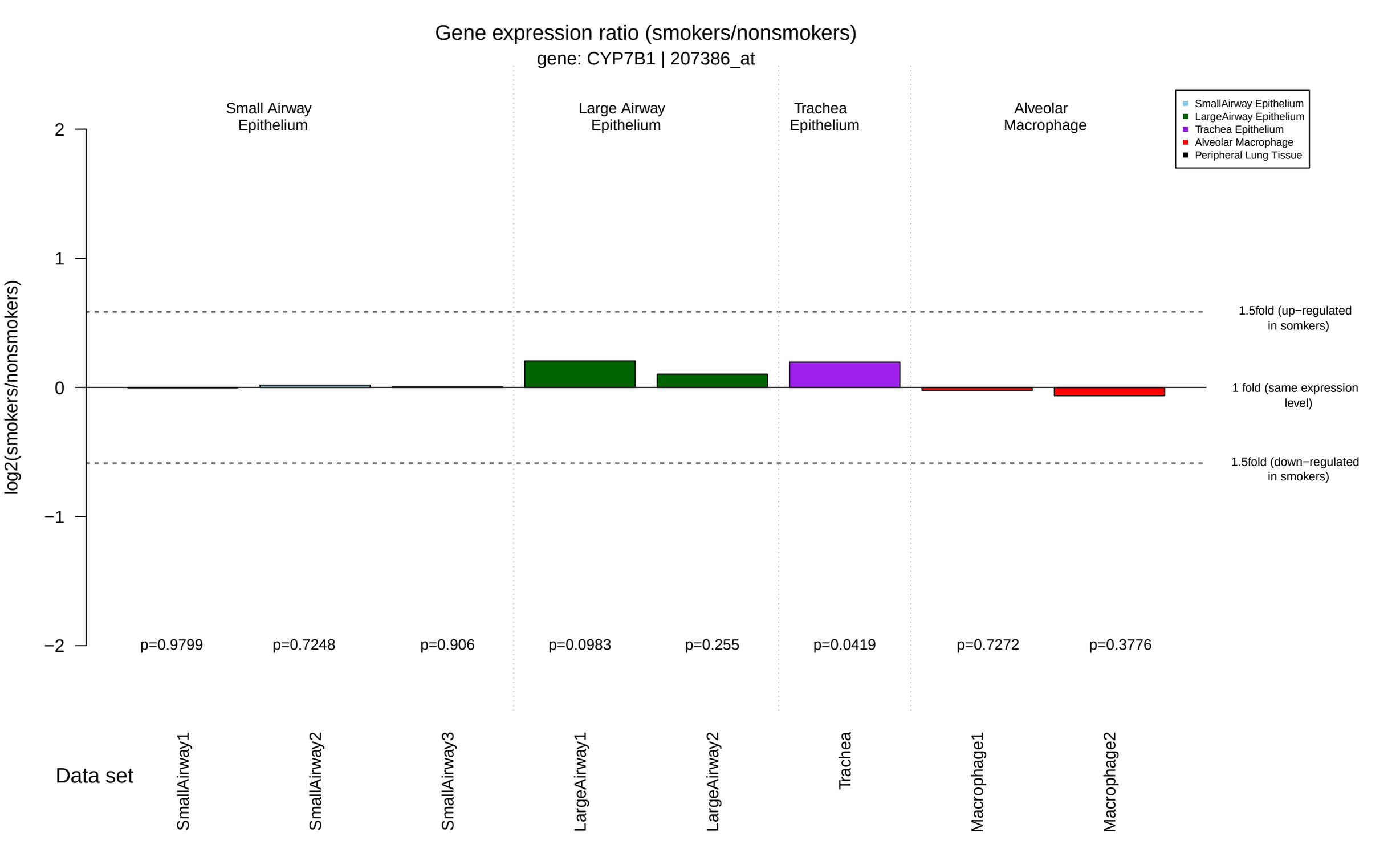
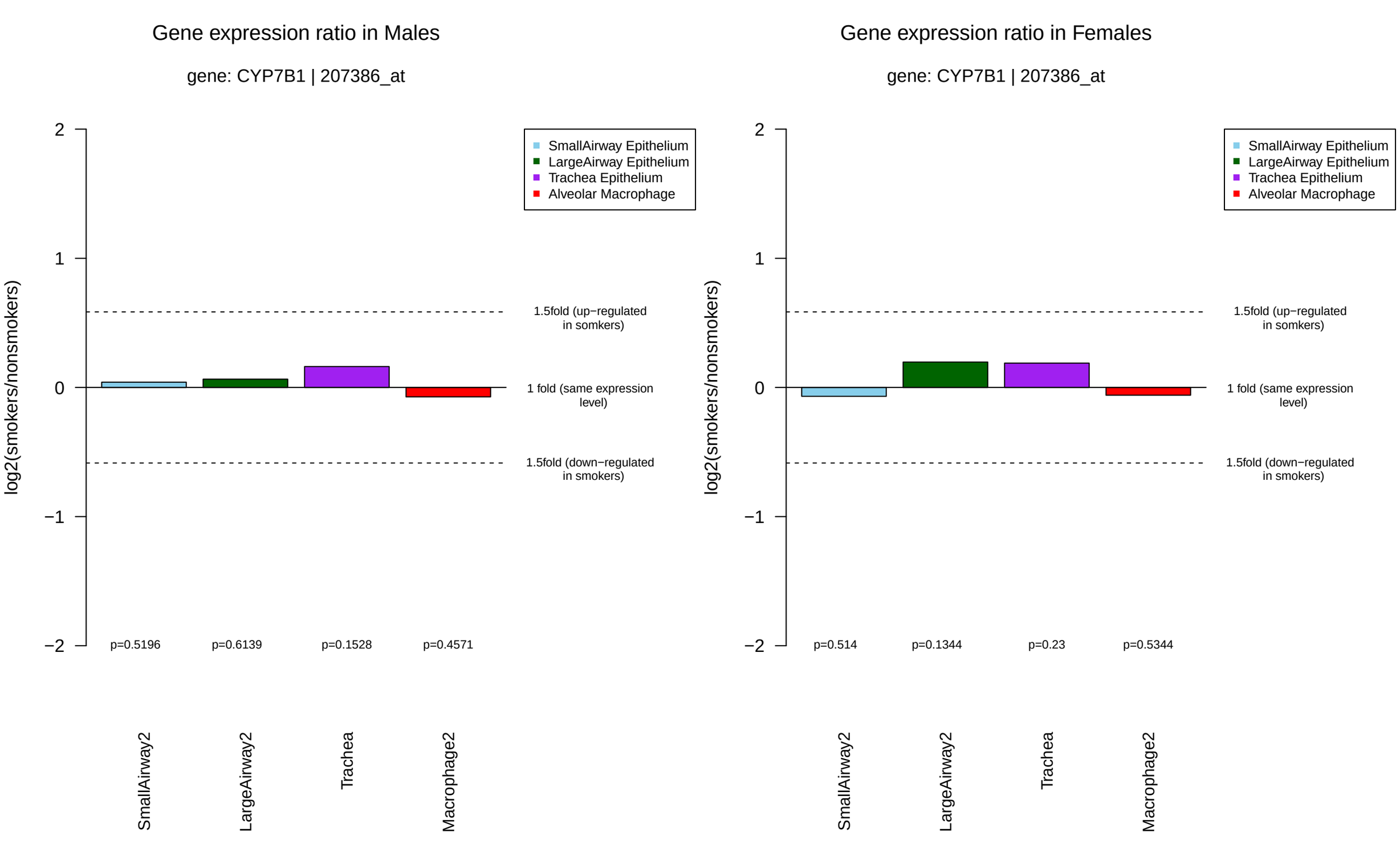
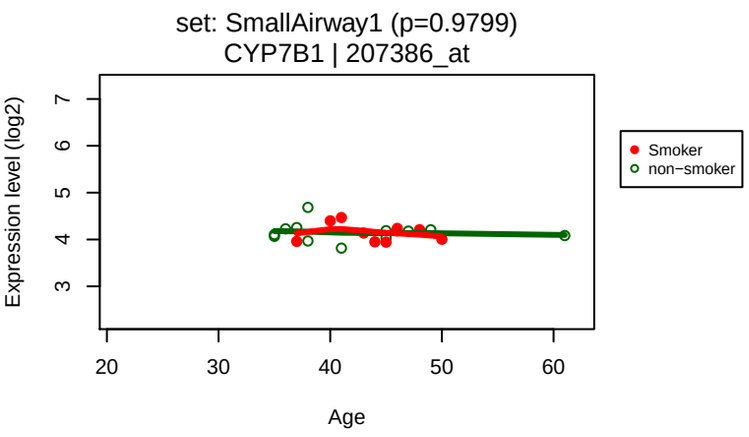
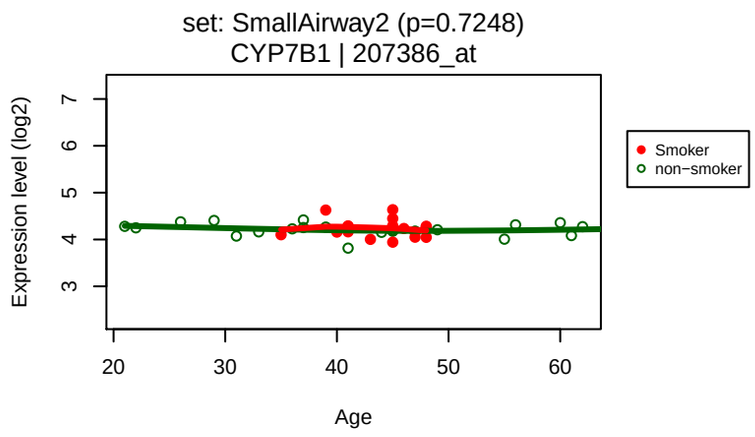
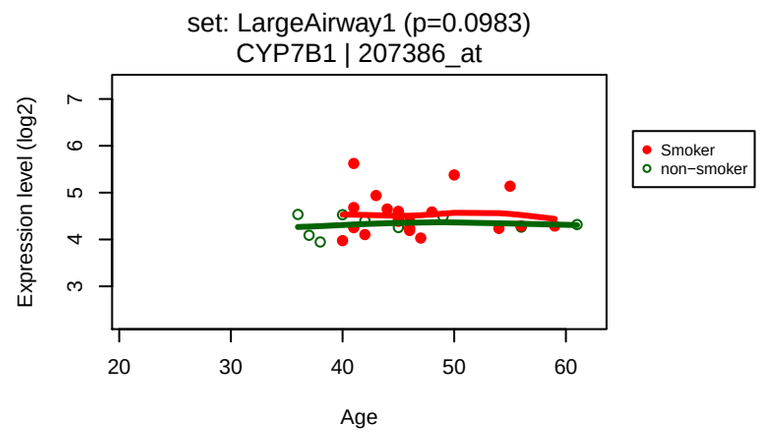

Gene expression of smokers and nonsmokers
gene: CYP7B1 | 207386_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9799
p=0.7248
p=0.906
p=0.0983
p=0.255
p=0.0419
p=0.7272
p=0.3776
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

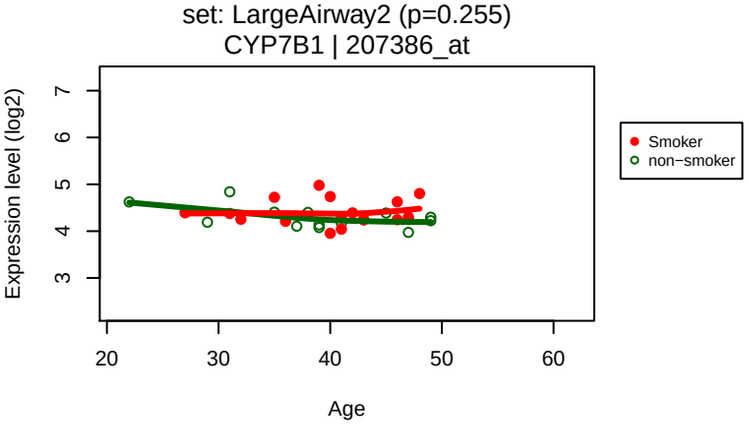

Gene expression ratio (smokers/nonsmokers)
gene: CYP7B1 | 207386_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9799
p=0.7248
p=0.906
p=0.0983
p=0.255
p=0.0419
p=0.7272
p=0.3776
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

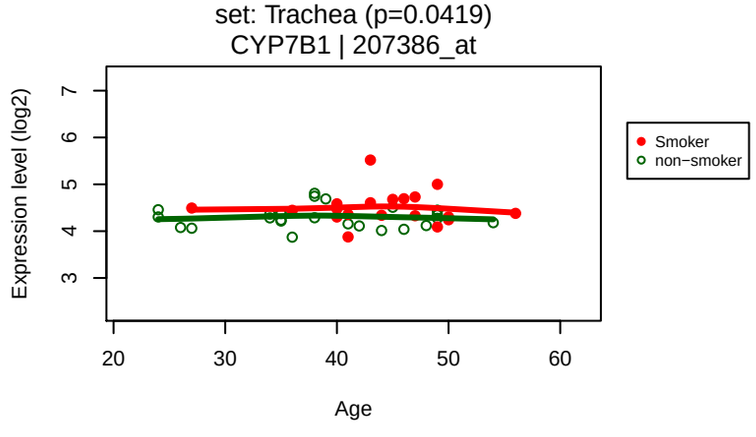

Gene expression ratio in Males
gene: CYP7B1 | 207386_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5196
p=0.6139
p=0.1528
p=0.4571
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP7B1 | 207386_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.514
p=0.1344
p=0.23
p=0.5344
SmallAirway2
LargeAirway2
Trachea
Macrophage2

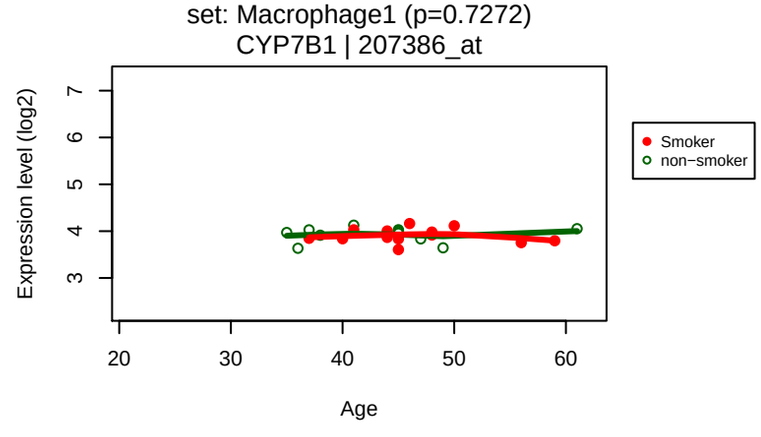

set: SmallAirway1 (p=0.9799)
CYP7B1 | 207386_at
Expression level (log2)
Age
Smoker
non−smoker

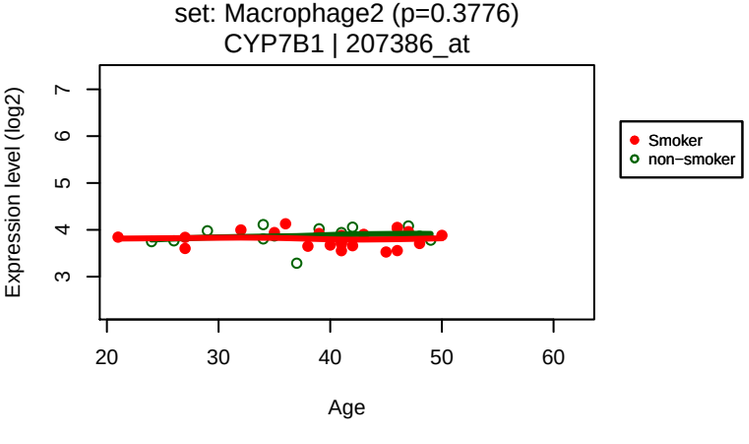

set: SmallAirway2 (p=0.7248)
CYP7B1 | 207386_at
Expression level (log2)
Age
Smoker
non−smoker

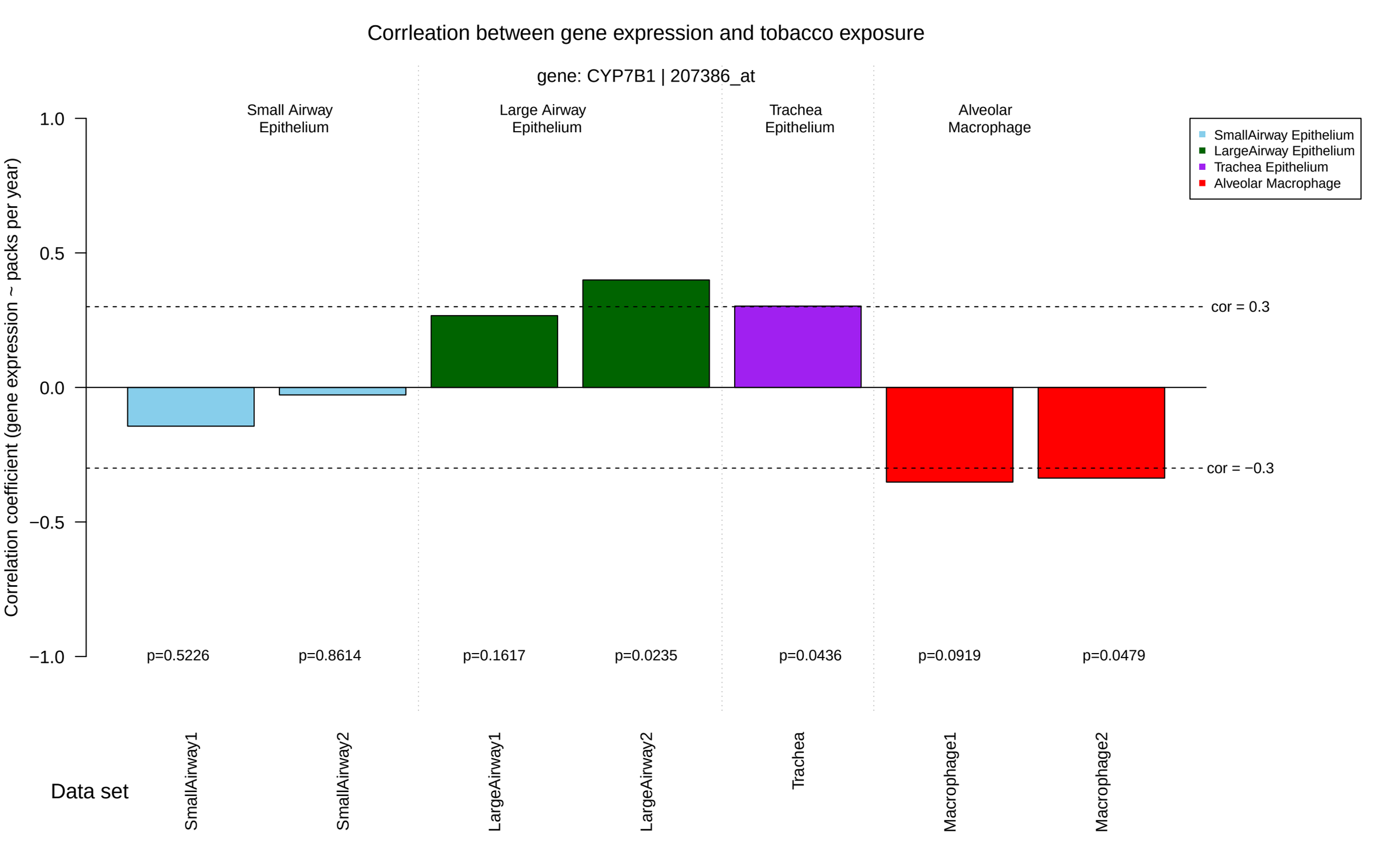

set: LargeAirway1 (p=0.0983)
CYP7B1 | 207386_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.255)
CYP7B1 | 207386_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0419)
CYP7B1 | 207386_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7272)
CYP7B1 | 207386_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3776)
CYP7B1 | 207386_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP7B1 | 207386_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5226
p=0.8614
p=0.1617
p=0.0235
p=0.0436
p=0.0919
p=0.0479
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2